Test Plan Document

Traffic Management System · Updated after Sprint 1 and Sprint 2 (Deliverable 3)
9 Test Cases — Unit, Integration and System Testing · executed 2026-11-12 11:52 IST

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01	Authentication Module	To test successful user login with valid credentials.	Registered user account and accessible login page.	1. Register the account. 2. Enter username and password. 3. Click Login. 4. Open the dashboard data.	Username: valid_user Password: Valid@123	User is authenticated and the dashboard is displayed.	200 "Login successful", role user; the token opens the traffic dashboard data. Automated in tests/test_test_plan.py (test_ut01_...)	**Pass**
UT-02	Authentication Module	To test login validation with invalid credentials.	Login page is accessible; registered account exists.	1. Open Login. 2. Enter an unknown username, then the real username with a wrong password. 3. Click Login.	invalid_user / wrong123 valid_user / wrong123	Login is rejected and a clear authentication error is displayed.	401 "Invalid credentials" in both cases; no token issued. Automated in tests/test_test_plan.py (test_ut02_...)	**Pass**
UT-03	Incident Reporting Module	To test submission of a valid traffic incident report.	User is authenticated and incident reporting is available.	1. Open Incident Reporting. 2. Select type. 3. Enter location and description. 4. Submit. 5. Reopen the incident list.	Type: Accident Location: MG Road Description: Two vehicles blocking lane	Input is accepted, the incident is stored, and a success message is displayed.	201 "Incident reported"; stored with the same type, location and description. Automated in tests/test_test_plan.py (test_ut03_...)	**Pass**
IT-01	Traffic Data & Live Map	To test integration between traffic data and the live map.	User is logged in; traffic data and map are available.	1. Set MG Road traffic to High. 2. Report an accident. 3. Load map data and alerts.	Road: MG Road Traffic level: High Incident: Accident	Traffic data is shown on the map with congestion and incident indicators.	MG Road returned as High with coordinates; Accident and Congestion alerts at MG Road. Automated in tests/test_test_plan.py (test_it01_...)	**Pass**
IT-02	Route Planning	To test integration of route planning with traffic data.	User is logged in; places and traffic data are available.	1. Request Majestic to Whitefield. 2. Set every road on that route (Route A) to High. 3. Request again and compare.	Source: Majestic Destination: Whitefield Traffic: High on Route A	System uses congestion data and shows the least-congested route.	Second route is congestion-aware and shares no road with Route A. Automated in tests/test_test_plan.py (test_it02_...)	**Pass**
IT-03	Alerts & Notifications	To test integration of traffic events with user alerts.	Two users subscribed: one to MG Road, one to Silk Board.	1. Report a road closure (Roadblock). 2. Check alerts. 3. Check both users' notifications.	Event: Road closure Area: MG Road	Relevant users receive a traffic alert about the road closure.	Closure alert at MG Road; MG Road subscriber notified; Silk Board subscriber not notified. Automated in tests/test_test_plan.py (test_it03_...)	**Pass**
ST-01	Traffic Management System	To test end-to-end traffic monitoring and route planning.	System running; user can log in; map and traffic data available.	1. Login. 2. Open live map. 3. Report accident. 4. Enter source/destination. 5. Request route. 6. View alerts.	Source: Majestic Destination: Whitefield Incident: Accident	User authenticated, traffic/map data shown, route calculated, relevant alerts displayed.	All steps succeed; route from Majestic to Whitefield with ETA; Accident alert shown; route saved in history. Automated in tests/test_test_plan.py (test_st01_...)	**Pass**
ST-02	Traffic Administration	To test administrator signal management with authorization and audit logging.	Administrator and a normal user exist; signal data exists.	1. Open Signal Management. 2. Select a signal. 3. Change status (also try as non-admin and with an invalid state). 4. Save. 5. Check audit log.	Signal: Majestic junction (plan: Junction-01) Old: RED New: GREEN	Authorised change is validated, stored, shown and recorded in the audit log.	Non-admin refused (403); invalid state refused (400); RED to GREEN saved (200); audit entry by admin1. Automated in tests/test_test_plan.py (test_st02_...)	**Pass**

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
ST-03	Emergency Vehicle Priority	To test end-to-end emergency priority and optimised routing.	Emergency operator logged in; heavy traffic on MG Road and Indiranagar.	1. Request emergency priority. 2. Select destination. 3. Receive optimised route. 4. Verify request log.	Vehicle: Ambulance-01 Destination: Whitefield (plan: City Hospital) Traffic: High	Authorised request accepted, optimised route provided, priority action logged.	201 accepted; emergency route uses live traffic and is no slower than the normal route; route signals cleared; request and audit entry stored. Automated in tests/test_test_plan.py (test_st03_...)	**Pass**

Run with `python3 -m pytest -q`. Result of the full suite: 276 tests, 275 passed, 0 failed, 1 skipped (Playwright browser module, not installed). Full reports: docs/reports/ (Unit Test, System Validation, Code Coverage, Branch Coverage).

Changes since the original Test Plan

Item	Change	Reason
All cases	Executed as automated pytest tests; actual results and Pass/Fail recorded	The original plan marked every case "Not executed — manual testing required"
UT-02	Also tries a real username with a wrong password	Both must give the same error so the message does not reveal which accounts exist (NFR-01)
IT-01	Checks the data the map draws: road level, coordinates, incident and congestion alerts	The map shows whatever the traffic and alerts services return
IT-02	"Route A" is the normal route; all its roads are set to High	Gives a clear pass rule instead of "a suitable route"
IT-03	Road closure is a Roadblock incident (raises a Closure alert); an unrelated user must not be notified	FR-11: only relevant users are notified
ST-02	Junction-01 is the Majestic signal; non-admin and invalid state are also tried	Signals are named after the junction they control; checks authorisation and validation
ST-03	City Hospital is replaced by Whitefield; "optimised" means traffic-aware and no slower than the normal route	Destinations must be places on the simulated network; emergency routes may use busy roads because their signals turn green

Test cases added during the sprints

Every SRS requirement now has automated test cases. Each row is one test file; every case is listed by name in docs/reports/Unit_Test_Report.pdf. Pass/Fail/Skip count test runs, so a test that runs once per input value counts once per run.

Requirement	Test file	Added	Pass	Fail	Skip	Requirement	Test file	Added	Pass	Fail	Skip
FR-01 Registration	test_register.py	Sprint 1	3	0	0	FR-18 Reports	test_reports.py	Sprint 2	11	0	0
FR-02 Login	test_login.py	Sprint 1	3	0	0	FR-19 Data management	test_storage.py	Sprint 1	4	0	0
FR-03 Role-based access	test_roles.py	Sprint 1	4	0	0	FR-19 Data management	test_data_management.py	Sprint 2	2	0	0
FR-04 Traffic monitoring	test_traffic.py	Sprint 1	3	0	0	FR-20 / NFR-12 Audit	test_audit.py	Sprint 2	12	0	0
FR-05 Live map	test_home.py	Sprint 1	1	0	0	NFR-01 Security	test_security.py	Sprint 1	3	0	0
FR-06/07 Signals	test_signals.py	Sprint 2	9	0	0	NFR-02 Response time	test_performance.py	Sprint 2	4	0	0
FR-08 Incident reporting	test_incidents.py	Sprint 1	3	0	0	NFR-03 Availability	test_availability.py	Sprint 2	6	0	0
FR-09 Incident management	test_incident_management.py	Sprint 2	8	0	0	NFR-04/08 Usability, browsers	test_compat_static.py	Sprint 2	10	0	0
FR-10 Alerts	test_alerts.py	Sprint 2	9	0	0	NFR-08 Real browsers	test_browsers.py	Sprint 2	0	0	1
FR-11 Notifications	test_notifications.py	Sprint 2	8	0	0	NFR-05 Validation	test_validation.py	Sprint 2	7	0	0
FR-12 Route planning	test_routing.py	Sprint 1	5	0	0	NFR-06 Scalability	test_scalability.py	Sprint 2	11	0	0
FR-13 Congestion routing	test_congestion_routing.py	Sprint 1	3	0	0	NFR-07 Modularity	test_modularity.py	Sprint 2	23	0	0
FR-14/15 Emergency	test_emergency.py	Sprint 2	12	0	0	NFR-09 Consistency	test_consistency.py	Sprint 2	25	0	0
FR-16 Parking	test_parking.py	Sprint 2	20	0	0	NFR-10 Privacy	test_privacy.py	Sprint 2	16	0	0
FR-17 Analytics	test_analytics.py	Sprint 2	8	0	0	NFR-11 Integration	test_integration_ready.py	Sprint 2	33	0	0